



# Operating Systems & Networks

## CTEC1704C

### Lecture – 14 : Transferring Files

Dr. Sameer Jha, SFHEA  
Faculty of Computing  
De Montfort University Cambodia



FTP, sFTP & SSH

*CTEC170*

*4*

*Networks*





## Lecture Content

- Transferring Files
  - FTP
    - Secure FTP
    - Alternative FTP protocols
  - SSH



## References & Reading

- [https://en.wikipedia.org/wiki/File\\_Transfer\\_Protocol](https://en.wikipedia.org/wiki/File_Transfer_Protocol)
- [ol http://www.faqs.org/rfcs/rfc114.html](http://www.faqs.org/rfcs/rfc114.html);  
[rfc959.html](http://www.faqs.org/rfcs/rfc959.html);
- [/rfc1579.html](http://www.faqs.org/rfcs/rfc1579.html); [rfc2428.html](http://www.faqs.org/rfcs/rfc2428.html); [rfc2822.html](http://www.faqs.org/rfcs/rfc2822.html)  
[erabilities.htm](https://sourcedaddy.com/networking/ftp-sharing-and-vulnerabilities.htm)
- [https://sourcedaddy.com/networking/ftp-sharing-and-vul](https://sourcedaddy.com/networking/ftp-sharing-and-vulnerabilities.htm)  
[https://en.wikipedia.org/wiki/Secure\\_She](https://en.wikipedia.org/wiki/Secure_Shell)  
[ll](https://en.wikipedia.org/wiki/Secure_Shell)
- [https://www.ssh.com/ssh/command/#sec-Using-the-Linu](https://www.ssh.com/ssh/command/#sec-Using-the-Linux-client)  
[X](https://www.ssh.com/ssh/command/#sec-Using-the-Linux-client)  
[-client](https://www.ssh.com/ssh/command/#sec-Using-the-Linux-client)

- in particular, watch the video



## Post lecture discussion topics

- Compare RFC 114 and RFC 959
  - what extra functionality is proposed in the later RFC?
- Read RFC 2428. What new technology does this RFC address and why?
- When is a passive FTP transfer mode used?
- sftp vs ssh???



## Lecture Content

- **Transferring Files**
  - Secure FTP
  - Alternative FTP protocols
  - SSH



## File Transfer Protocol

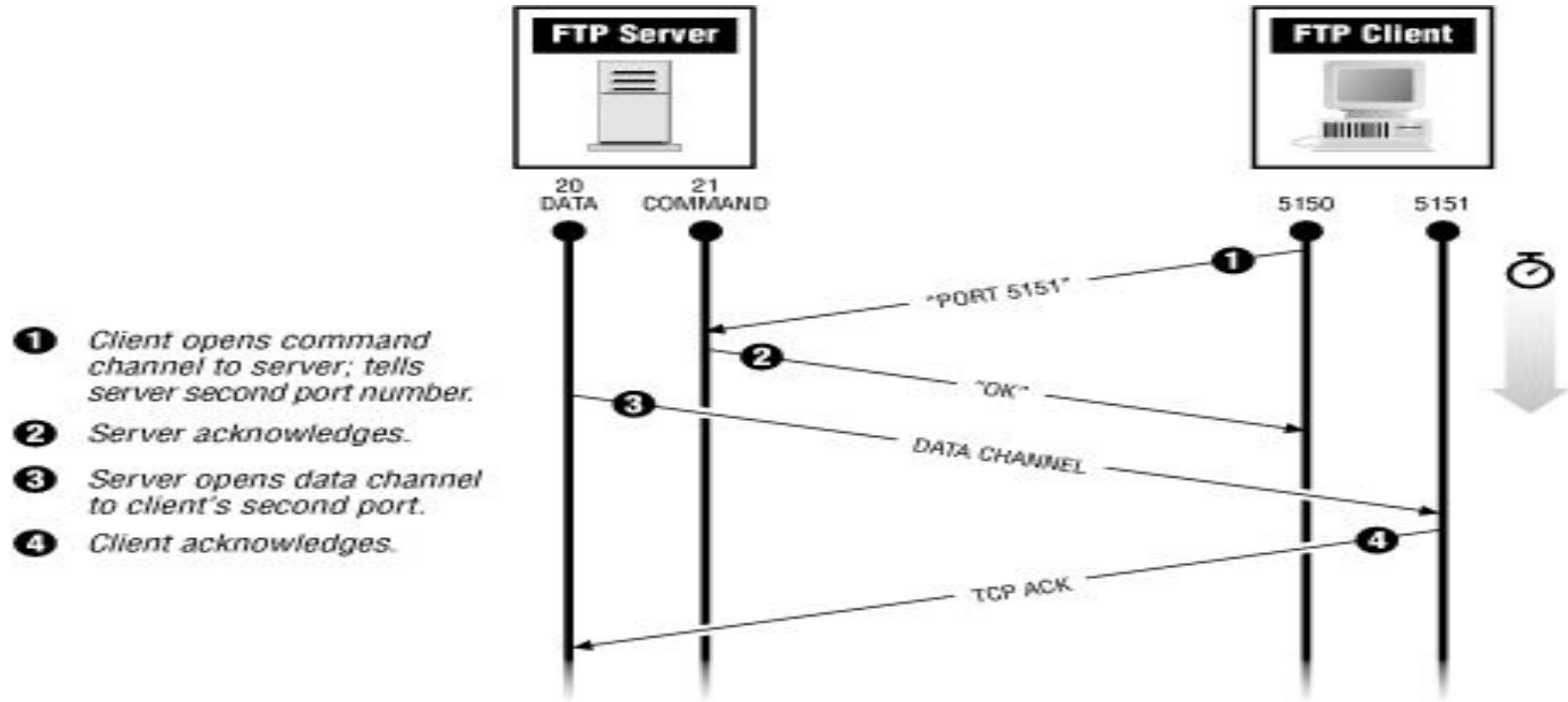
- Shares files and directories between remote systems
  - reliable and efficient data transfer
- Uses TCP/IP protocols
- Telnet protocol
  - client-server architecture
  - can be secured with SSL/TLS
  - may implement simple authentication
- Uses 2 ports:
  - 21 – command
  - 20 – data



- FTP is defined in:
  - RFC 114 (16 April 1971)
  - RFC 765 (June 1980)
  - RFC 959 (October 1985)
  - RFC 2228 (June 1997)
  - RFC 2428 (September 1998)



## FTP Ports





## Mode

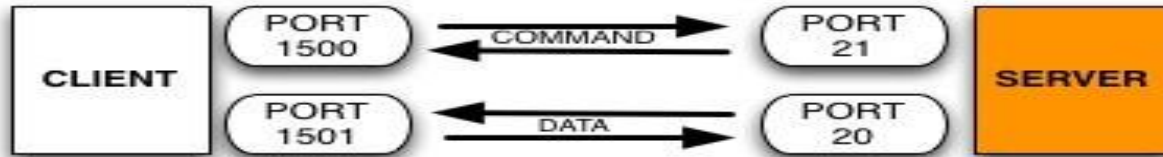
- Active or passive mode
  - determines how the data connection is established



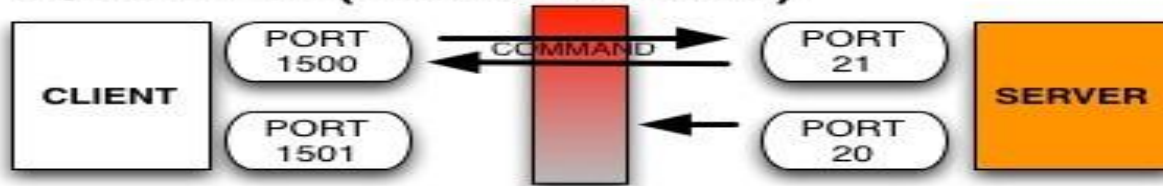
## Active Mode

- The client creates a TCP control connection

### ACTIVE FTP



### ACTIVE FTP (WITH FIREWALL)

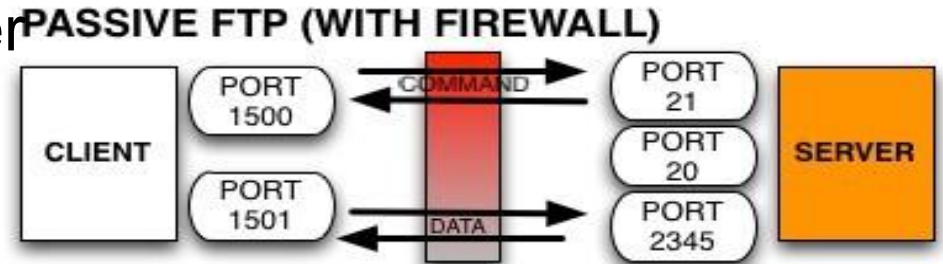


The firewall blocks the server's attempt to communicate with the client because the server uses a different port than the first connection.



## Active Mode

- Clients behind firewalls are unable to accept incoming TCP connections
- Client sends a PASV command
  - receives a server IP address
  - server port number



The firewall doesn't block the server's attempt to communicate with the client because the client initiated the communication both times.



## Data Representation

- ASCII mode:
  - used for text
- Image mode:
  - the sending machine sends file byte by byte
- EBCDIC mode:
  - IBM text
- Local mode:
  - identical machines can use proprietary format



## Data Transfer Modes

- Stream mode
  - data is sent as a continuous stream
- Block mode
  - FTP breaks the data into several blocks
- Compressed mode
  - data is compressed using a single algorithm



- Normal
  - username and password
    - . USER command
    - . PASS command
- Anonymous login
  - server configuration
  - login in with eg
  - email

often used for software



## NAT & Firewalls

Connections initiated from Internet not allowed NATs use internal (non-routeable) addresses

- Solutions
  - can use PASV
  - NAT alters PORT numbers

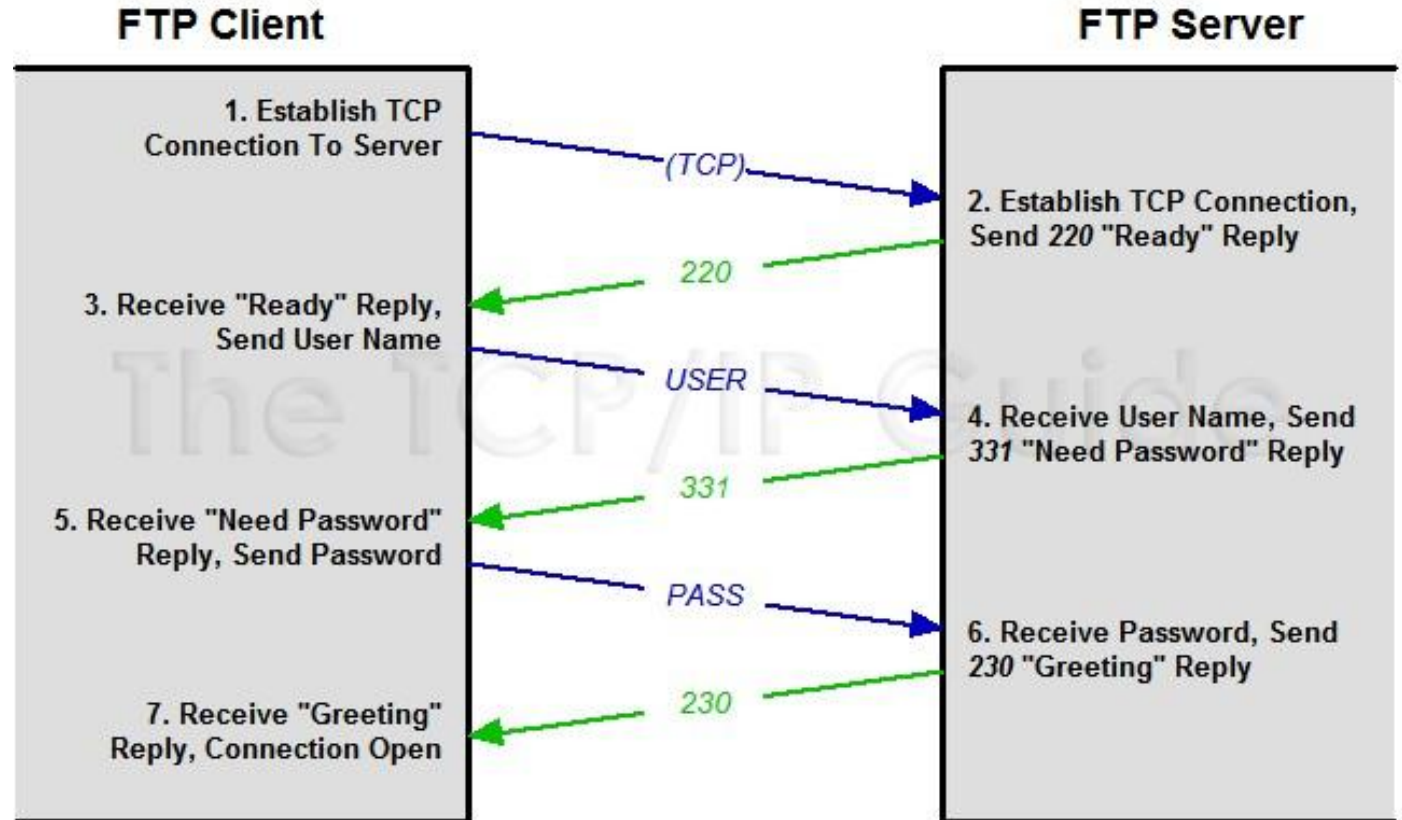




## Browsers

- Most browsers can initiate FTP downloads
- Plugins can give upload facility

## Message flow





Apply a display filter ... <Ctrl-/>

Expression...

| No. | Time      | Source          | Destination     | Protocol | Length | Info                                                  |
|-----|-----------|-----------------|-----------------|----------|--------|-------------------------------------------------------|
| 4   | 0.000226  | 192.168.150.200 | 192.168.150.100 | TCP      | 74     | 21 → 59671 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS= |
| 5   | 0.000450  | 192.168.150.100 | 192.168.150.200 | TCP      | 66     | 59671 → 21 [ACK] Seq=1 Ack=1 Win=5856 Len=0 TSval=52  |
| 6   | 0.065553  | 192.168.150.200 | 192.168.150.100 | FTP      | 126    | Response: 220 ProFTPD 1.3.1 Server (Debian) [::ffff:  |
| 7   | 0.065824  | 192.168.150.100 | 192.168.150.200 | TCP      | 66     | 59671 → 21 [ACK] Seq=1 Ack=61 Win=5856 Len=0 TSval=5  |
| 8   | 10.468735 | 192.168.150.100 | 192.168.150.200 | FTP      | 77     | Request: USER bill                                    |
| 9   | 10.468854 | 192.168.150.200 | 192.168.150.100 | TCP      | 66     | 21 → 59671 [ACK] Seq=61 Ack=12 Win=5792 Len=0 TSval=  |
| 10  | 10.475396 | 192.168.150.200 | 192.168.150.100 | FTP      | 98     | Response: 331 Password required for bill              |
| 11  | 10.476119 | 192.168.150.100 | 192.168.150.200 | TCP      | 66     | 59671 → 21 [ACK] Seq=12 Ack=93 Win=5856 Len=0 TSval=  |
| 12  | 14.171841 | 192.168.150.100 | 192.168.150.200 | FTP      | 81     | Request: PASS password                                |
| 13  | 14.235921 | 192.168.150.200 | 192.168.150.100 | FTP      | 92     | Response: 230-Welcome to my world!                    |
| 14  | 14.236696 | 192.168.150.100 | 192.168.150.200 | TCP      | 66     | 59671 → 21 [ACK] Seq=27 Ack=119 Win=5856 Len=0 TSval= |
| 15  | 14.236919 | 192.168.150.200 | 192.168.150.100 | FTP      | 91     | Response: 230 User bill logged in                     |
| 16  | 14.237052 | 192.168.150.100 | 192.168.150.200 | TCP      | 66     | 59671 → 21 [ACK] Seq=27 Ack=144 Win=5856 Len=0 TSval= |
| 17  | 14.237850 | 192.168.150.100 | 192.168.150.200 | FTP      | 72     | Request: SYST                                         |
| 18  | 14.238117 | 192.168.150.200 | 192.168.150.100 | FTP      | 85     | Response: 215 UNIX Type: L8                           |
| 19  | 14.266431 | 192.168.150.100 | 192.168.150.200 | TCP      | 66     | 59671 → 21 [ACK] Seq=33 Ack=163 Win=5856 Len=0 TSval= |
| 20  | 36.877768 | 192.168.150.100 | 192.168.150.200 | FTP      | 72     | Request: QUIT                                         |
| 21  | 36.878527 | 192.168.150.200 | 192.168.150.100 | FTP      | 80     | Response: 221 Goodbye.                                |
| 22  | 36.879276 | 192.168.150.200 | 192.168.150.100 | TCP      | 66     | 21 → 59671 [FIN, ACK] Seq=177 Ack=39 Win=5792 Len=0   |
| 23  | 36.880929 | 192.168.150.100 | 192.168.150.200 | TCP      | 66     | 59671 → 21 [ACK] Seq=39 Ack=177 Win=5856 Len=0 TSval= |
| 24  | 36.881604 | 192.168.150.100 | 192.168.150.200 | TCP      | 66     | 59671 → 21 [FIN, ACK] Seq=39 Ack=178 Win=5856 Len=0   |
| 25  | 36.881631 | 192.168.150.200 | 192.168.150.100 | TCP      | 66     | 21 → 59671 [ACK] Seq=178 Ack=40 Win=5792 Len=0 TSval= |
| 26  | 39.225657 | 192.168.150.100 | 192.168.150.200 | TCP      | 74     | 59672 → 21 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK   |
| 27  | 39.225675 | 192.168.150.200 | 192.168.150.100 | TCP      | 74     | 21 → 59672 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS= |

- ▶ Frame 3: 74 bytes on wire (592 bits), 74 bytes captured (592 bits)
- ▶ Ethernet II, Src: 26:04:4f:79:64:58 (26:04:4f:79:64:58), Dst: 2e:ac:d1:86:5b:36 (2e:ac:d1:86:5b:36)
- ▶ Internet Protocol Version 4, Src: 192.168.150.100, Dst: 192.168.150.200
- ▶ Transmission Control Protocol, Src Port: 59671, Dst Port: 21, Seq: 0, Len: 0

0000 2e ac d1 86 5b 36 26 04 4f 79 64 58 08 00 45 00 ...[6&.. oydX..E.  
0010 00 3c 00 e4 40 00 40 06 8b 5a c0 a8 96 64 c0 a8 ..<..@.@..Z...d..  
0020 96 c8 e9 17 00 15 6d 2b 14 99 00 00 00 00 a0 02 .....m+ .....  
0030 16 d0 03 46 00 00 02 04 05 b4 04 02 08 0a 00 00 ...F.....





|                 |           |                                                                                                  |
|-----------------|-----------|--------------------------------------------------------------------------------------------------|
| 192.168.150.200 | FTP       | 96 Request: PORT 192,168,150,100,185,168                                                         |
| 192.168.150.100 | FTP       | 95 Response: 200 PORT command successful                                                         |
| 192.168.150.200 | FTP       | 81 Request: STOR text.txt                                                                        |
| 192.168.150.100 | TCP       | 74 20 → 47528 [SYN] Seq=0 Win=5840 Len=0 MSS=1460 SACK_PERM=1 TSval=49165 TSecr=0 WS=32          |
| 192.168.150.200 | TCP       | 74 47528 → 20 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=1460 SACK_PERM=1 TSval=25375 TSecr=491.. |
| 192.168.150.100 | TCP       | 66 20 → 47528 [ACK] Seq=1 Ack=1 Win=5856 Len=0 TSval=49165 TSecr=25375                           |
| 192.168.150.100 | FTP       | 120 Response: 150 Opening BINARY mode data connection for text.txt                               |
| 192.168.150.200 | FTP-DA... | 133 FTP Data: 67 bytes (PORT) (STOR text.txt)                                                    |
| 192.168.150.100 | TCP       | 66 20 → 47528 [ACK] Seq=1 Ack=68 Win=5856 Len=0 TSval=49165 TSecr=25375                          |
| 192.168.150.200 | TCP       | 66 47528 → 20 [FIN, ACK] Seq=68 Ack=1 Win=5792 Len=0 TSval=25375 TSecr=49165                     |
| 192.168.150.100 | TCP       | 66 20 → 47528 [FIN, ACK] Seq=1 Ack=69 Win=5856 Len=0 TSval=49165 TSecr=25375                     |
| 192.168.150.200 | TCP       | 66 47528 → 20 [ACK] Seq=69 Ack=2 Win=5792 Len=0 TSval=25375 TSecr=49165                          |
| 192.168.150.200 | TCP       | 66 59674 → 21 [ACK] Seq=86 Ack=265 Win=5856 Len=0 TSval=25379 TSecr=49165                        |
| 192.168.150.100 | FTP       | 89 Response: 226 Transfer complete                                                               |
| 192.168.150.200 | TCP       | 66 59674 → 21 [ACK] Seq=86 Ack=288 Win=5856 Len=0 TSval=25379 TSecr=49168                        |

|              |                 |                 |     |                                                         |
|--------------|-----------------|-----------------|-----|---------------------------------------------------------|
| 59 63.073553 | 192.168.150.200 | 192.168.150.100 | FTP | 91 Response: 230 User bill logged in                    |
| 60 63.073896 | 192.168.150.100 | 192.168.150.200 | TCP | 66 59673 → 21 [ACK] Seq=27 Ack=144 Win=5856 Len=0 TSval |
| 61 63.074098 | 192.168.150.100 | 192.168.150.200 | FTP | 72 Request: SYST                                        |
| 62 63.076427 | 192.168.150.200 | 192.168.150.100 | FTP | 85 Response: 215 UNIX Type: L8                          |
| 63 63.116915 | 192.168.150.100 | 192.168.150.200 | TCP | 66 59673 → 21 [ACK] Seq=33 Ack=163 Win=5856 Len=0 TSval |
| 64 81.256223 | 192.168.150.100 | 192.168.150.200 | FTP | 71 Request: PWD                                         |
| 65 81.257114 | 192.168.150.200 | 192.168.150.100 | FTP | 109 Response: 257 "/home/bill" is the current directory |
| 66 81.257283 | 192.168.150.100 | 192.168.150.200 | TCP | 66 59673 → 21 [ACK] Seq=38 Ack=206 Win=5856 Len=0 TSval |
| 67 84.600989 | 192.168.150.100 | 192.168.150.200 | FTP | 96 Request: PORT 192,168,150,100,132,131                |
| 68 84.602478 | 192.168.150.200 | 192.168.150.100 | FTP | 95 Response: 200 PORT command successful                |
| 69 84.603197 | 192.168.150.100 | 192.168.150.200 | TCP | 66 59673 → 21 [ACK] Seq=68 Ack=235 Win=5856 Len=0 TSval |
| 70 84.603202 | 192.168.150.100 | 192.168.150.200 | FTP | 72 Request: LIST                                        |



|    |           |                 |                 |     |                                                          |
|----|-----------|-----------------|-----------------|-----|----------------------------------------------------------|
| 29 | 39.242993 | 192.168.150.200 | 192.168.150.100 | FTP | 126 Response: 220 ProFTPD 1.3.1 Server (Debian) [::ffff: |
| 30 | 39.243246 | 192.168.150.100 | 192.168.150.200 | TCP | 66 59672 → 21 [ACK] Seq=1 Ack=61 Win=5856 Len=0 TSval=9  |
| 31 | 45.136751 | 192.168.150.100 | 192.168.150.200 | FTP | 78 Request: USER bilge                                   |
| 32 | 45.136781 | 192.168.150.200 | 192.168.150.100 | TCP | 66 21 → 59672 [ACK] Seq=61 Ack=13 Win=5792 Len=0 TSval=  |
| 33 | 45.146275 | 192.168.150.200 | 192.168.150.100 | FTP | 99 Response: 331 Password required for bilge             |
| 34 | 45.146901 | 192.168.150.100 | 192.168.150.200 | TCP | 66 59672 → 21 [ACK] Seq=13 Ack=94 Win=5856 Len=0 TSval=  |
| 35 | 48.089568 | 192.168.150.100 | 192.168.150.200 | FTP | 78 Request: PASS james                                   |
| 36 | 48.092258 | 192.168.150.200 | 192.168.150.100 | FTP | 88 Response: 530 Login incorrect.                        |
| 37 | 48.092732 | 192.168.150.100 | 192.168.150.200 | TCP | 66 59672 → 21 [ACK] Seq=25 Ack=116 Win=5856 Len=0 TSval= |
| 38 | 48.093093 | 192.168.150.100 | 192.168.150.200 | FTP | 72 Request: SYST                                         |
| 39 | 48.093507 | 192.168.150.200 | 192.168.150.100 | FTP | 85 Response: 215 UNIX Type: L8                           |
| 40 | 48.136682 | 192.168.150.100 | 192.168.150.200 | TCP | 66 59672 → 21 [ACK] Seq=31 Ack=135 Win=5856 Len=0 TSval= |
| 41 | 54.373213 | 192.168.150.100 | 192.168.150.200 | FTP | 72 Request: QUIT                                         |
| 42 | 54.373719 | 192.168.150.200 | 192.168.150.100 | FTP | 80 Response: 221 Goodbye.                                |



## FTP Vulnerabilities

- Brute force attacks
- Bounce attacks
- Packet capture (sniffing)
- Port stealing
- Spoof attacks
- Username protection



## Lecture Content

- Transferring Files
  - FTP
  - **Secure FTP**
  - Alternative FTP protocols
  - SSH



## Secure FTP

- Secure transmission
- Encrypts username and password
- Encrypts the content
- SSL/TLS ("FTPS")
- SSH File Transfer Protocol ("SFTP") may be used instead
  - technologically different





- Not related to FTP
- Uses Secure Shell (SSH) to transfer files
- Encrypts both commands and data
  - uses an SSL tunnel
- Functionally similar to Standard FTP
- SFTP clients cannot talk to an SFTP server
  - and vice versa



## DMU account access

- Connect to H: account
  - sftp.dmu.ac.uk
  - ssh.dmu.ac.u
- k sFTP/SSH
  - clientS - FileZilla
  - Linux - FileZilla
  - Windows – BitVise
  - Client either - PuTTY
- First time of access you may be asked to accept the certificate



- Transferring Files
  - FTP
    - Secure FTP
    - **Alternative FTP protocols**
  - SSH



## Trivial File Transfer Protocol

- Uses UDP on port 69
- Small and easy to implement
- No user authentication
- Fixed block size
- Stop and wait protocol
- Often used when booting from LAN
  - PXE boot



## Explicit FTPS

- Extension to the FTP standard
- Clients can request that the FTP session be encrypted
  - AUTH TLS
- Server has the option of allowing or denying connections that do not request TLS



## Implicit FTPS

- Deprecated standard for FTP
- Required the use of a SSL or TLS
- Used different ports than plain FTP



## FTP over SSH

- Not SFTP
- Practice of tunnelling a normal FTP session over an SSH connection
  - difficult because FTP uses multiple TCP connections
- Most SSH clients
  - protect the command channel
  - but ignore the data channel



- Transferring Files
  - FTP
    - Secure FTP
    - Alternative FTP protocols
- **SSH**





## Secure Shell

- Cryptographic network protocol for operating network services securely over an unsecured network
- uses public-key cryptography to authenticate the remote computer and allow it to authenticate the user, if necessary
- Typical applications
  - remote command-line login
  - remote command execution
- Any network service can be secured with SSH
- Files leaked by Edward Snowden indicate that the National Security Agency can sometimes decrypt SSH
  - allowing them to read the contents of SSH sessions



## SSH - File transfer protocols

- Secure copy (SCP)
  - evolved from RCP protocol over SSH
- rsync
  - intended to be more efficient than SCP
  - generally runs over an SSH connection

### SSH File Transfer Protocol (SFTP)

secure alternative to FTP (not to be confused with FTP over SSH or FTPS)

- Files transferred over shell protocol (a.k.a. FISH)
  - released in 1998, evolved from Unix shell commands over SSH
- Fast and Secure Protocol (FASP)
  - aka Aspera



## Using SSH

- Default port
  - 22
- ssh [user@999.999.999.999](#)
- ssh [user@999.999.999.999](#) -P 2222
- sftp [user@999.999.999.999](#) -p 2222



Apply a display filter ... &lt;Ctrl-/&gt;

Expression... +

| No. | Time      | Source          | Destination     | Protocol | Length | Info                                                   |
|-----|-----------|-----------------|-----------------|----------|--------|--------------------------------------------------------|
| 13  | 37.132434 | 192.168.150.120 | 192.168.150.110 | TCP      | 74     | 22 → 40828 [SYN, ACK] Seq=0 Ack=1 Win=5792 Len=0 MSS=  |
| 14  | 37.132679 | 192.168.150.110 | 192.168.150.120 | TCP      | 66     | 40828 → 22 [ACK] Seq=1 Ack=1 Win=5856 Len=0 TSval=42   |
| 15  | 37.232718 | 192.168.150.120 | 192.168.150.110 | SSHv2    | 98     | Server: Protocol (SSH-2.0-OpenSSH_5.1p1 Debian-4)      |
| 16  | 37.232851 | 192.168.150.110 | 192.168.150.120 | TCP      | 66     | 40828 → 22 [ACK] Seq=1 Ack=33 Win=5856 Len=0 TSval=4   |
| 17  | 37.233297 | 192.168.150.110 | 192.168.150.120 | SSHv2    | 98     | Client: Protocol (SSH-2.0-OpenSSH_5.1p1 Debian-4)      |
| 18  | 37.233303 | 192.168.150.120 | 192.168.150.110 | TCP      | 66     | 22 → 40828 [ACK] Seq=33 Ack=33 Win=5792 Len=0 TSval=   |
| 19  | 37.233659 | 192.168.150.110 | 192.168.150.120 | SSHv2    | 858    | Client: Key Exchange Init                              |
| 20  | 37.233664 | 192.168.150.120 | 192.168.150.110 | TCP      | 66     | 22 → 40828 [ACK] Seq=33 Ack=825 Win=7392 Len=0 TSval=  |
| 21  | 37.242748 | 192.168.150.120 | 192.168.150.110 | SSHv2    | 850    | Server: Key Exchange Init                              |
| 22  | 37.243002 | 192.168.150.110 | 192.168.150.120 | SSHv2    | 90     | Client: Diffie-Hellman Group Exchange Request          |
| 23  | 37.258762 | 192.168.150.120 | 192.168.150.110 | TCP      | 66     | 22 → 40828 [ACK] Seq=817 Ack=849 Win=7392 Len=0 TSval= |
| 24  | 37.278952 | 192.168.150.120 | 192.168.150.110 | SSHv2    | 218    | Server: Diffie-Hellman Group Exchange Group            |
| 25  | 37.280120 | 192.168.150.110 | 192.168.150.120 | SSHv2    | 210    | Client: Diffie-Hellman Group Exchange Init             |
| 26  | 37.280129 | 192.168.150.120 | 192.168.150.110 | TCP      | 66     | 22 → 40828 [ACK] Seq=969 Ack=993 Win=8960 Len=0 TSval= |
| 27  | 37.287516 | 192.168.150.120 | 192.168.150.110 | SSHv2    | 786    | Server: Diffie-Hellman Group Exchange Reply, New Key   |
| 28  | 37.309343 | 192.168.150.110 | 192.168.150.120 | TCP      | 66     | 40828 → 22 [ACK] Seq=993 Ack=1689 Win=10560 Len=0 TS   |
| 29  | 46.086665 | 192.168.150.110 | 192.168.150.120 | SSHv2    | 82     | Client: New Keys                                       |
| 30  | 46.116546 | 192.168.150.120 | 192.168.150.110 | TCP      | 66     | 22 → 40828 [ACK] Seq=1689 Ack=1009 Win=8960 Len=0 TS   |
| 31  | 46.116885 | 192.168.150.110 | 192.168.150.120 | SSHv2    | 114    | Client: Encrypted packet (len=48)                      |
| 32  | 46.116899 | 192.168.150.120 | 192.168.150.110 | TCP      | 66     | 22 → 40828 [ACK] Seq=1689 Ack=1057 Win=8960 Len=0 TS   |

Checksum: 0xb81f [unverified]

[Checksum Status: Unverified]

Urgent pointer: 0

Options: (24 bytes), No-Operation (NOP), No-Operation (NOP), Timestamps, No-Operation (NOP), No-Operation (NOP), SACK

[SEQ/ACK analysis]

```

0000  26 c0 35 85 4a b0 c2 79 2f 72 c9 11 08 00 45 10  &5.J.y /r...E.
0010  00 40 0f 9b 40 00 40 06 7c d5 c0 a8 96 6e c0 a8  .@...@. |...n..
0020  96 78 9f 7d 00 16 dc a0 de db dd 0b bc 7c b0 10  .x... .....
```



- Transferring Files
  - FTP
    - Secure FTP
    - Alternative FTP protocols
  - SSH

Q&A

*Thank You!*



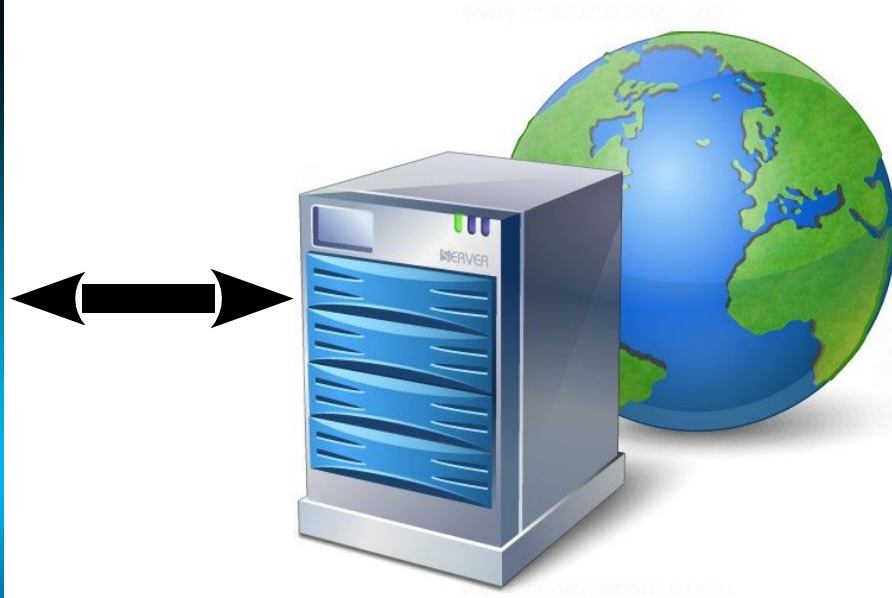
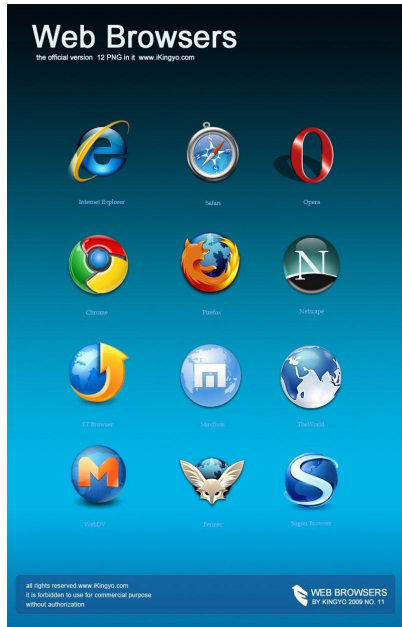
# Operating Systems & Networks

## CTEC1704C

### Lecture – 15 : Web Client & Servers

Dr. Sameer Jha, SFHEA  
Faculty of Computing  
De Montfort University Cambodia





*Web Clients &*

- Who runs the Internet?
- Web Clients
  - Web Servers
  - Hosting & Server Farms
  - Security & Privacy Issues

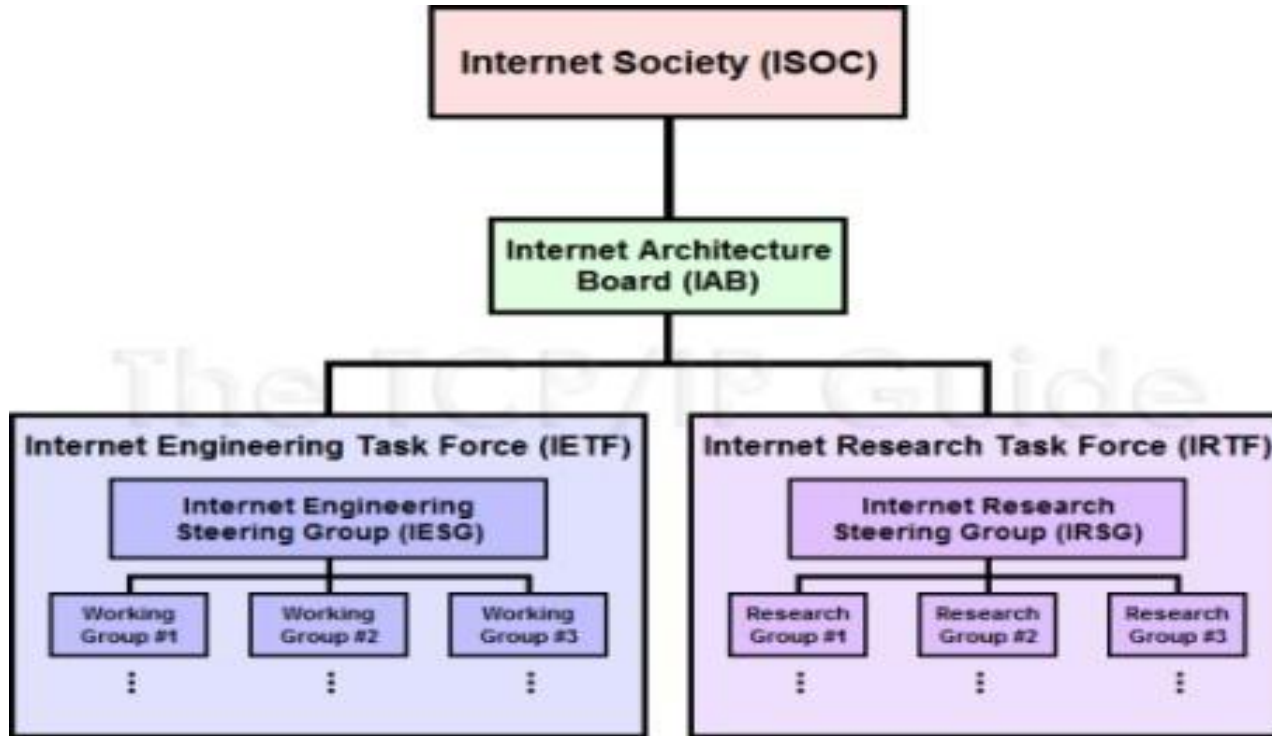
## References & Reading

- [https://en.wikipedia.org/wiki/Web\\_browser](https://en.wikipedia.org/wiki/Web_browser)
- [https://en.wikipedia.org/wiki/Web\\_server](https://en.wikipedia.org/wiki/Web_server)
- <https://html.spec.whatwg.org/multipage/>
- [https://en.wikipedia.org/wiki/Document\\_Object\\_Model](https://en.wikipedia.org/wiki/Document_Object_Model)
- <http://blog.kaspersky.com/browser-security-features-mozilla-firefox/>
- <http://www.hardwareanalysis.com/content/article/1549/>
- <http://www.seoconsultants.com/articles/1000/cache-control>
- [http://www.what-is-what.com/what\\_is/apache.html](http://www.what-is-what.com/what_is/apache.html)
- <https://en.wikipedia.org/wiki/Nginx>
- <https://www.hiawatha-webserver.org/>
- [https://en.wikipedia.org/wiki/Internet\\_Information\\_Services](https://en.wikipedia.org/wiki/Internet_Information_Services)
- <https://www.torproject.org/> and <https://proton.me/>
- <https://duckduckgo.com/>

- Explain how media types assist a browser to correctly display a web object
- Describe how the DOM is used to support the rendering of a web page
- What are the main security issues caused by JS?

- **Who runs the Internet?**
  - Web Clients
  - Web Servers
- Hosting & Server Farms
- Security & Privacy Issues

The Internet is run by...?





## The Internet is run by...?

- Internet Society (ISOC)
- Internet Architecture Board (IAB)
- Internet Engineering Steering Group (IESG)
- Internet Engineering Task Force (IETF)
- Internet Research Task Force (IRTF)
- Internet Corporation for Assigned Names and Numbers (ICANN)
  - Internet Assigned Numbers Authority (IANA)
    - Regional Internet Registry (RIR)

The Internet is run by...?

- Application layer:
  - W3C (World Wide Web Consortium)
- Physical layer
  - IEEE 802 standards



- Who runs the Internet?
- **Web Clients**
- Web Servers
- Hosting & Server Farms
- Security & Privacy Issues



# Web Clients

- Primary function: retrieving, presenting and traversing information resources on WWW
- Information resource is identified by a Uniform Resource Identifier
- Web Objects
- Hyperlinks
  - Doug Englebart, Ted Nelson



# Web Clients

- Google Chrome, Mozilla Firefox, Internet Edge (Explorer), Opera, and Safari
- Telnet
- Multiple protocols
  - http, https, ftp, sftp, file, mailto, news, etc
- Implement HTTP GET & POST methods
- Extension plugins process known file types
  - unknown file types are offered for saving

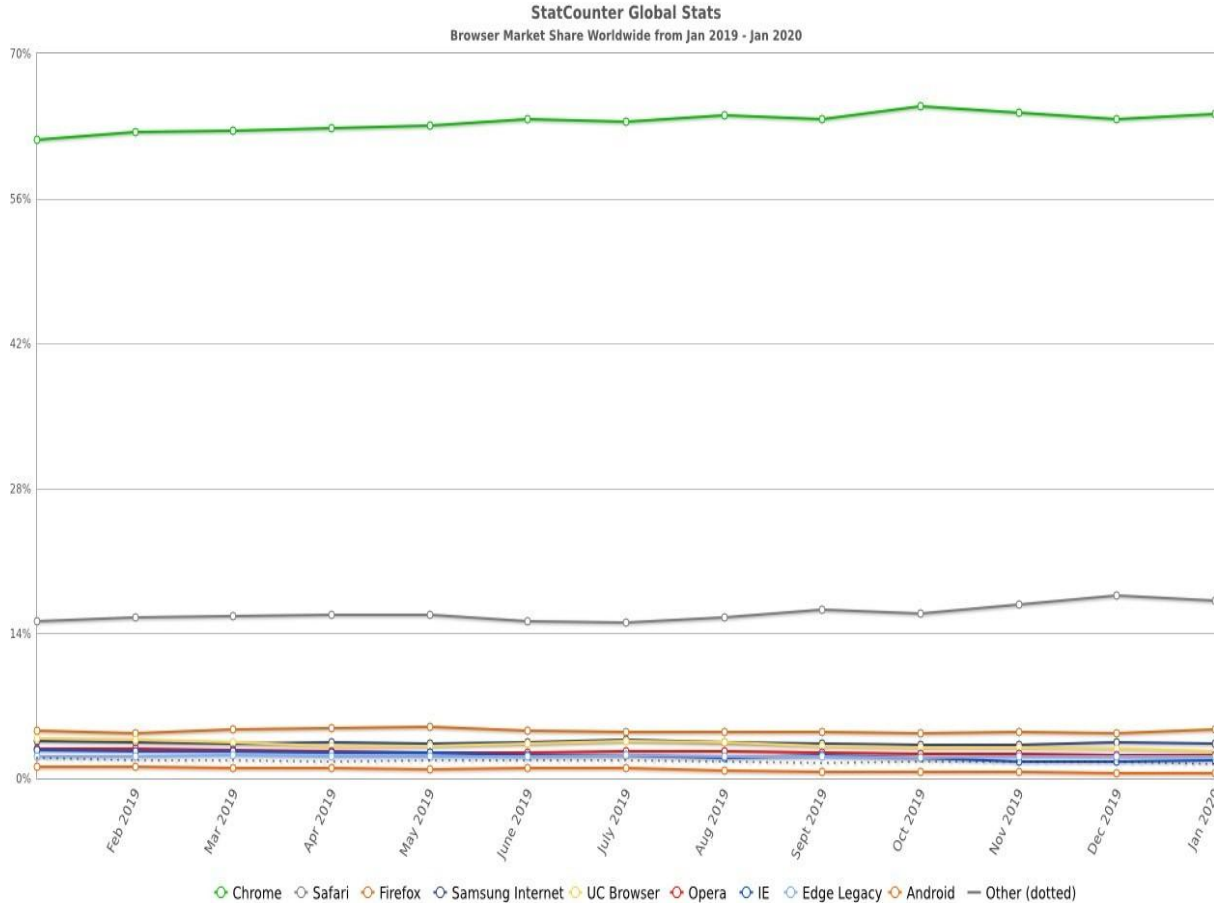
# Browser Features

- . Page Navigation
- . Search within page
- Security & Privacy
- Support non-standard HTML

# Browser Components

- . User Interface
- . Layout Engine
- . Rendering Engine
- . JavaScript interpreter
- . UI backend
- . Networking component
- . Data persistence component
- . Resolver

# Market Share



# HTML(5)

| HTML                                                                                                                                                                                                                                                          | xhtml                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>&lt;!doctype html&gt; &lt;html lang="en"&gt;   &lt;head&gt;     &lt;meta charset="UTF-8"&gt;     &lt;title&gt;Example document&lt;/title&gt;   &lt;/head&gt;   &lt;body&gt;     &lt;p&gt;Example paragraph&lt;/p&gt;   &lt;/body&gt; &lt;/html&gt;</pre> | <pre>&lt;?xml version="1.0" encoding="UTF-8"?&gt; &lt;html xmlns="<a href="http://www.w3.org/1999/xhtml">http://www.w3.org/1999/xhtml</a> " &gt;   &lt;head&gt;     &lt;title&gt;Example document&lt;/title&gt;   &lt;/head&gt;   &lt;body&gt;     &lt;p&gt;Example paragraph&lt;/p&gt;   &lt;/body&gt; &lt;/html&gt;</pre> |



# DOM Model

- Cross-platform and language-independent
- Representing and interacting with convention
- objects Objects in the DOM tree may be addressed and manipulated by using methods on the objects
  - Using JavaScript/JScript



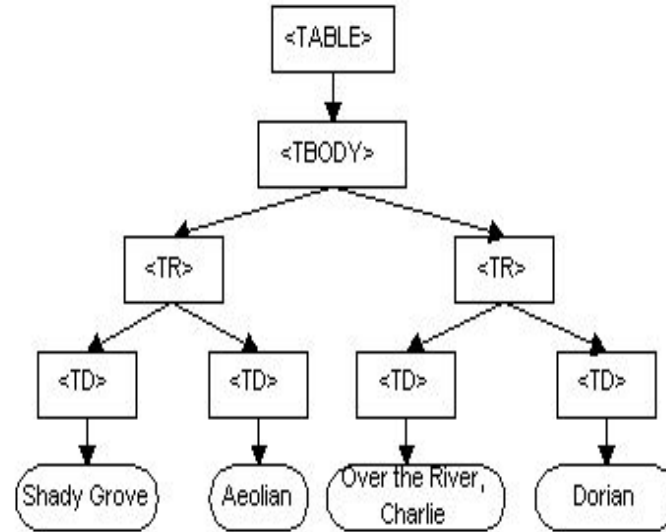


# DOM

- Most browsers implement a DOM
- Tree structure
  - Top node = Document object
- HTML is parsed into a DOM tree

## Converting the HTML to a DOM

```
<TABLE>
<TBODY>
<TR>
<TD>Shady Grove</TD>
<TD>Aeolian</TD>
</TR>
<TR>
<TD>Over the River,
Charlie</TD>
<TD>Dorian</TD>
</TR>
</TBODY>
</TABLE>
```



- Cascading Style Sheet
- Technical specifications for a layout
- Style settings “cascade” to the lowest node of the DOM tree
- Always a default CSS

- Two-part identifier for file formats
- Originally SMTP attachments
- Non-ASCII message components
- Parts: A type, a subtype, and zero or more optional parameters
  - eg text/html; charset=UTF-8
- Can define custom types

# JavaScript



- . Object-based interpreted language
- . Client-side scripts
  - interact with the user
  - control the browser
  - communicate asynchronously
  - alter displayed document content

- Node.js
- ECMAScript language standard

# JS Libraries

- JQuery
- Dojo
- Facilitate easy integration with other technologies eg Ruby, PHP, CSS, etc
- Open licence



# Security & Privacy Issues

# Security & Privacy

- . Privacy Settings
- . Firefox add-ons
- Search tools



The screenshot shows the Tor Project's homepage. The browser window title is "Tor Project: Anonymity Online - Tor Browser". The address bar shows "https://www.torproject.org/index.html.en". The page features a navigation menu with links: Home, About Tor, Documentation, Press, Blog, and Contact. Below the navigation menu are buttons for Download, Volunteer, and Donate. The main content area is divided into several sections:

- Anonymity Online**: A green box with the text "Protect your privacy. Defend yourself against network surveillance and traffic analysis." and a "Download Tor" button.
- What is Tor?**: A section explaining that Tor is free software and an open network that helps users defend against traffic analysis, network surveillance, and threats to personal freedom and privacy.
- Why Anonymity Matters**: A section explaining that Tor protects users by bouncing their communications around a distributed network of relays run by volunteers.
- Recent Blog Posts**: A list of recent blog posts, including "Tor Weekly News — October 15th...", "New SSLv3 attack found: Disable...", "Tor Weekly News — October 8th, 2014", "A closer look at the Great Fire...", and "Tor Weekly News — October 1st, 2014".
- Who Uses Tor?**: A section with three sub-sections: "Family & Friends", "Businesses", and "Activists".
- Our Projects**: A section with two sub-sections: "Tails" (Live CD/USB operating system) and "Orbot" (Tor for Google Android devices).

- . Who runs the Internet?
- . Web Clients
  - JavaScript
  - Security & Privacy Issues

- Web Servers
  - Hardware
  - Software
- Hosting & Server Farms



- What does optimising a web server involve?
- What features would you normally expect to find that a web server implements?
- List and briefly explain the most common forms of attack against a web server



- **Web Servers**
  - Hardware
  - Software
- **Hosting & Server Farms**



# Web Servers

- . Primary function: deliver web pages
- . Uses HTTP & HTTPS
- User agent initiates contact,  
requesting content
  - Request-response cycle of page requests



# Web Servers

- Supplementary function: receive content from clients
- Support server-side scripting languages
  - static content vs dynamic content

# Web Servers

- Embedded in devices

- printers

- routers

- light

- fittings

- loaders

- 

- camera

- s etc

IoT devices

# Common Features

- Virtual hosting
  - serve many web sites using one IP address
- Bandwidth throttling
  - limit speed of responses
  - prevent saturating the network
    - serve more clients
- Server-side scripting
  - generate dynamic web pages
  - separate web server and website



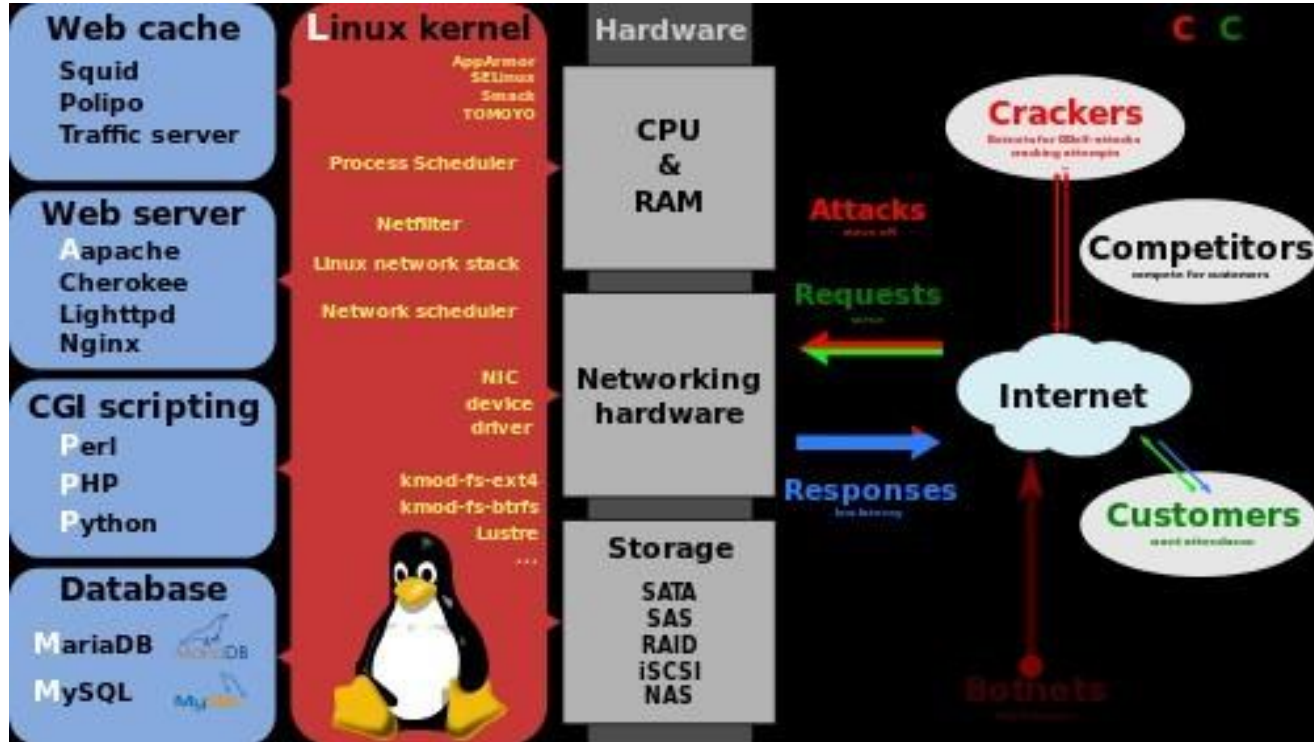
# Path Translation

- Map the path component of a Uniform Resource Locator (URL)
  - local file system resource
    - static requests
  - internal or external program name
    - dynamic requests

# Static pages

- URL
  - <http://www.example.com/path/file.html>
- HTTP 1.1 request:
  - GET /path/file.html HTTP/1.1
  - Host: [www.example.com](http://www.example.com)
- [www.example.com](http://www.example.com) maps to the home directory:
  - Apache: /var/www/html
  - IIS:
    - drive:\inetpub\wwwroot
- /var/www/html/path/file.html

# Lamp Stack



## Load limits

- Can handle only a limited number of concurrent client connections
  - between 2 and 80,000 per IP address
- Can serve a max requests per second
  - configuration settings
  - HTTP request type
  - static or dynamic
  - content cached content
  - host's hardware and software
- When a web server is near to or over its limits, it becomes unresponsive

# Overload Causes

- Too much legitimate concurrent web traffic
- Distributed Denial of Service attacks
- Computer worms that sometimes cause abnormal traffic
- XSS viruses can cause high traffic Internet
- bots
- Internet (network) slowdowns
- Web servers (computers) partial





# Anti-Overload Techniques

- Managing network traffic, by using:
  - Firewalls, HTTP traffic managers, bandwidth management and traffic shaping
- Web cache techniques
- Using different domain names to serve different content
- Using different domain names and/or hosts to separate different sized big files
- Multiple web servers (programs) per host



# Anti-Overload Techniques

- Using a load balancer
- Adding more hardware resources
- Tuning OS parameters for hardware capabilities and usage
- Using more efficient computer programs for web servers, etc.
- Using other workarounds, especially if dynamic content is involved



- Web Servers
  - **Hardware**
  - Software
- Hosting & Server Farms



# Optimal Server Configuration

- Need sub-second response times
  - Different metrics available:
    - total web page latency
    - time to first byte

# Server Bottlenecks

- HD access
  - Disk latency
  - Database requests
- Servers are not CPU intensive, so can eg
  - optimise code
  - compress output



# Caching

- RAM is cheap
    - maximise capacity
    - use cache software
  - 80 - 20 rule
    - 80% of traffic is for 20% of web pages
- "Formulating a caching policy for a web site - determining which resources should not get cached, which resources should, and for how long - is a vital step in improving site performance."

# Hardware Configuration

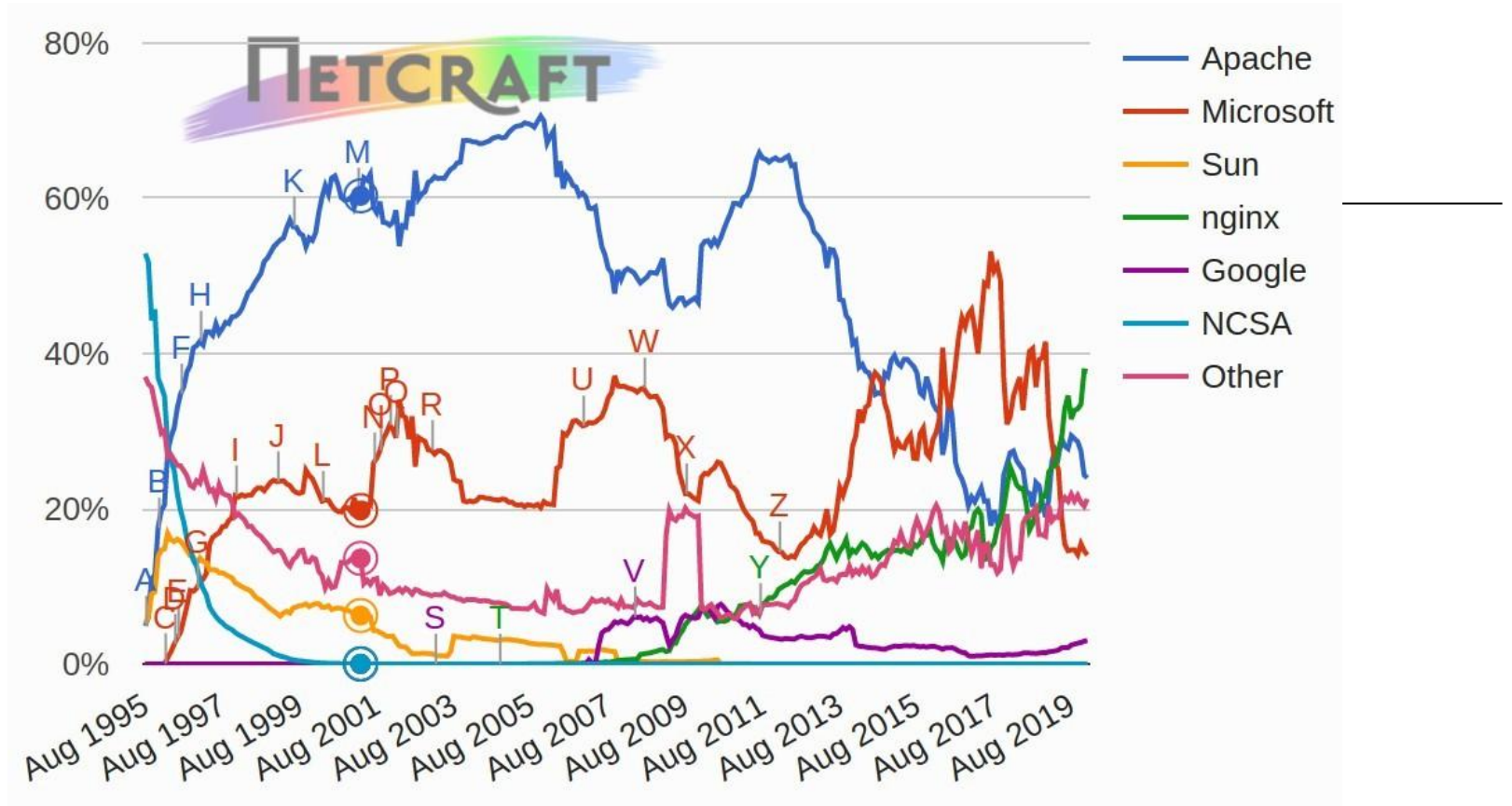
- . Maximum RAM
- . RAID
  - Multiple HD
- . Fast HD/SSD
- . Multiple NIC
- . Fast processor
  - 64-bit
  - large multiple cores

- Web Servers
  - Hardware
  - **Software**
- Hosting & Server Farms



# Web Server Daemons

- . Apache
- . IIS
- . Nginx
- . Google
- (Lighttpd, Zeus, Jigsaw, Hiawatha)





# Apache HTTP Server

- Open source
  - 501c non-profit organisation
- Multi-platform
- Services ~ 30% of web sites
- Foundation manages > 50 projects
- Multiple extension modules
- eg PHP



# nginx HTTP Server

- Open source reverse proxy server
  - HTTP, HTTPS
  - SMTP, POP3, IMAP
  - load balancer
  - HTTP cache
  - web server
- Multi-platform
- Asynchronous event-driven

# Hiawatha Web Server

- Open source webserver
- Focus on security
- Started in January 2002
- Small size
  - Good for embedded systems
- Simple configuration

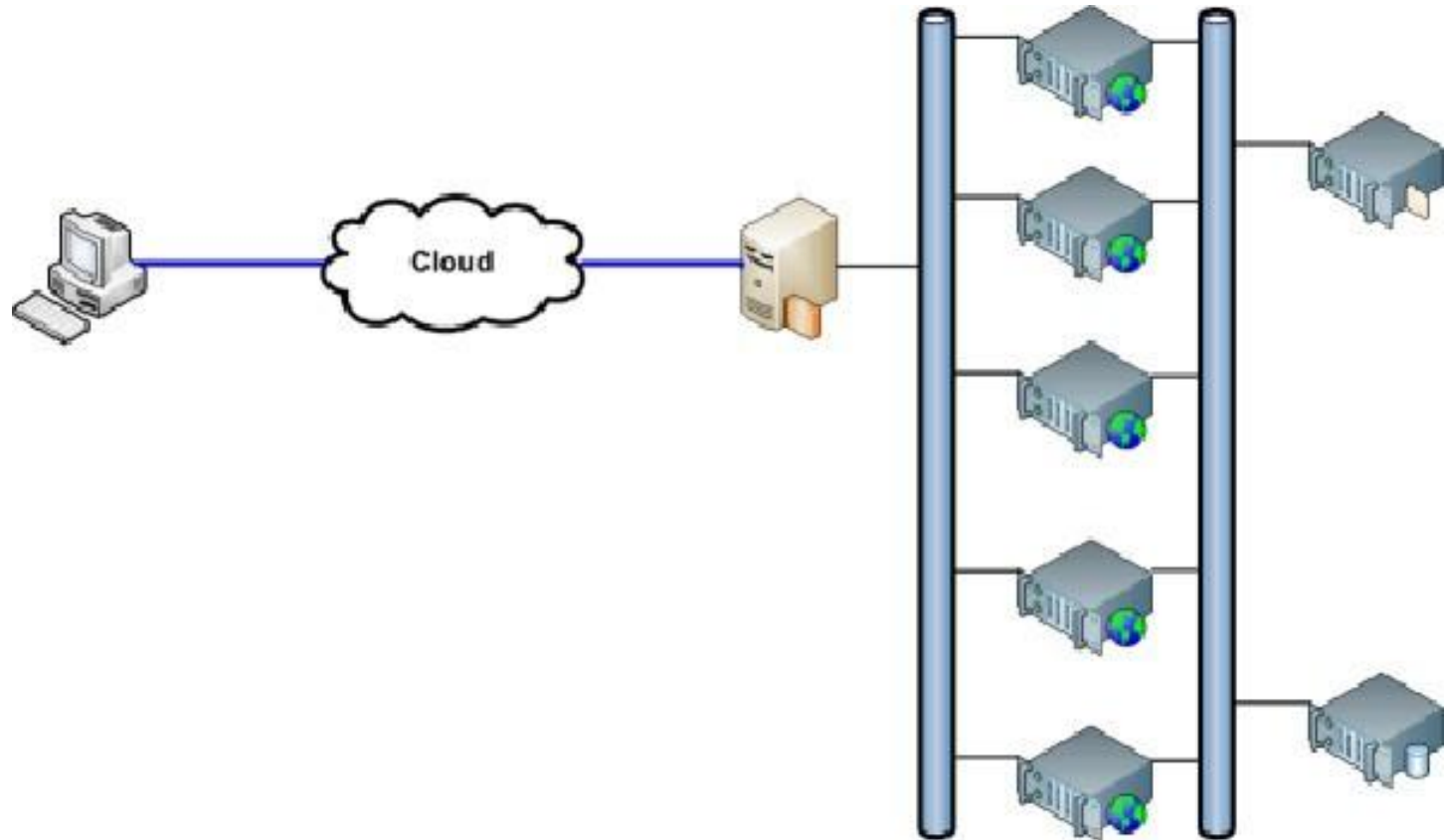
# IIS/Microsoft

- Proprietary
- HTTP, HTTPS, FTP, FTPS, SMTP and NNTP
- Modular architecture
  - Security modules
  - Content modules
  - Compression modules
  - Caching modules
  - Logging and Diagnostics modules

# Hosting & Server Farms

- Collection of computer servers
- Supply server functionality far beyond the capability of a single machine
  - Can be thousands of computers
- Expensive
  - financial
  - environmental





- Web Servers
  - Hardware
  - Software
- Hosting & Server Farms

Q&A

*Thank You!*